DraftFM: Zero-Shot Drafting of Unseen *Magic: The Gathering* Sets from Public Card Features

Brian Ward
Independent Researcher

July 9, 2026

Abstract

Draft assistants for *Magic: The Gathering* need weeks of post-release human pick data before they can model a new set, leaving them nothing to offer on release day. We train DraftFM, a pick model that represents every card as a frozen vector of public information, structured Scryfall features plus a sentence embedding of its rules text, with no card or set specific parameters, so it can score a set the moment its card list goes public. Trained on 149.4M picks from 28 17Lands sets, DraftFM agrees with high-win-rate players' picks 54.3% of the time on three sets it never trained on: 78.6% of what a model trained directly on the same set achieves, by our pre-registered normalized score, and above every other published number we could find that a real day-one deployment could have used, each drawn from a different test set and so a reference point rather than a head-to-head comparison. A simpler architecture, dropping the component that reads the set's full card list, scored no worse within seed noise, so we deployed it instead: this recipe, F-full, has 1.6M parameters and produces the paper's headline result, a single pre-registered pass over the first public snapshot of MSH, a licensed-IP set untouched by any training or tuning in this pipeline, reaching 57.0% agreement with high-win-rate players' picks. We release the code, the frozen evaluation protocol, model weights, and per-pick prediction files.

Contents

Table 1.1: Published pick-agreement results for *Magic: The Gathering* drafting, and whether each approach can score a previously unseen set using only information available on release day. The table closes with this paper's frozen Marvel Super Heroes (MSH) result. An asterisk marks a caveat: the method uses post-release information, or possible pretraining exposure to post-release discussion cannot be ruled out.

Reference	Approach	Unseen	Day 1	Test set	Top-1 (%)
Ward et al. (2021)	Hand-tuned ratings	no	yes	M19 (Draftsim)	44.5
Ward et al. (2021)	One-hot MLP, per set	no	no	M19 (Draftsim)	48.7
Brooks (2024)	One-hot MLP, per set	no	no	per set (within-set)	70.0
Czerner (2025)	Transformer, per set	no	no	FDN-era (within-set)	71.0
Bertram et al. (2024a)	Card features, one set	yes	yes	unseen (NEO-trained)	33.6–35.6
Bertram et al. (2024a)*	Features+stats+images, 13 sets	yes	no	BRO (unseen)	55.4
Bertram (2025)*	LLM, zero-shot	yes	no	NEO	43.0
DraftFM (this paper)	Card features, 31 sets	yes	yes	MSH (unseen)	57.0

1 Introduction

Drafting in *Magic: The Gathering* is a sequential game of imperfect information. Eight players open packs of cards and pass them around the table, each privately assembling a deck one pick at a time. Every few months a new set of several hundred cards is released, and drafters must learn it from scratch. For a data-driven assistant this creates a day-one problem: supervised pick models train on public draft logs, and those logs appear on a delay of roughly two weeks after a set's release (Brooks, 2024). The players who most want help with a new set are the ones no trained model can yet serve.

On release day, the state of the art is human judgment. Draftsim publishes hand-tuned ratings written by human experts (Draftsim, 2026), and drafters consult them because nothing learned is available. The learned within-set literature begins with Ward et al. (2021), who compare drafting agents on Core Set 2019 (M19): a random agent takes the human's card 22% of the time, a bot driven by expert-tuned ratings 44.5%, and a neural network trained on human drafts 48.7%. Its successors define the deployed state of the art. The statistical-drafting models of Brooks (2024), multilayer perceptrons trained per set, reach roughly 70% pick agreement, and the puder transformer of Czerner (2025) reports 71%. The statistical-drafting architecture is also the direct ancestor of the per-set ceiling models we measure against in Section 5. All of these models encode cards as vocabulary indices. Each learns from picks made in the one set it serves, so none can score an unseen set, and none can exist until the two-week data lag has passed. Table 1.1 collects these published results and records which of them a deployment could have used on release day.

Large language models look like a way around the lag, because they can read a new card's rules text the day it is published. Bertram (2025) test this directly. Zero-shot GPT-4o agrees with human picks 43% of the time on Kamigawa: Neon Dynasty (NEO), below the expert-tuned ratings above, and supplying each card's full rules text in the prompt lowers its accuracy further. Open 7–8B models fail to pick legal cards at all until LoRA-tuned on a million picks from the target set, after which they still trail small supervised multilayer perceptrons at far higher inference cost. Even the 43% figure carries a caveat. NEO was released years before GPT-4o's training cutoff, so post-release discussion of the set's strategy cannot be ruled out of the pretraining corpus, a leak a frozen feature-only encoder cannot have. The design lesson we take is to use a frozen text embedding as a feature, not a language model as the policy.

This paper asks how far pure transfer can go. DraftFM is a single pick model trained across every public 17Lands draft set (17Lands, 2026). To DraftFM, a card is nothing but a frozen vector of public information: structured features parsed from its Scryfall record (Scryfall, 2026), plus a sentence embedding of its rules text. The model has no set-identity parameters and no per-card weights. Whatever it knows about a set it must infer from the cards in the pack and the drafter's accumulating pool, and that inference transfers to a set released tomorrow, which a memorized card vocabulary cannot.

The nearest prior work is by Bertram et al. (2024a), who established the problem of drafting a set the model never trained on. They replace one-hot card identities with concatenated feature vectors: hand-engineered attributes, a Sentence-BERT embedding of card text (Reimers and Gurevych, 2019), a learned autoencoding of card images, and sixteen post-release usage statistics from 17Lands. Two of their findings shape our design. Feature representations match one-hot accuracy within a set, so nothing is lost by generalising, and multi-set pretraining, not the representation alone, is what buys transfer to an unseen set. Pretrained on thirteen sets and evaluated on held-out The Brothers' War (BRO), their best representation reaches 55.4% pick agreement. But that representation includes the usage-statistics block, which is computed from human play after a set's release, so 55.4% is an anchor for generalising across cards rather than for day-one deployment. Their day-one-feasible representation, features only, reaches 33.6–35.6% on unseen sets under single-set training, and no multi-set, features-only number was published. Filling that gap is this paper's headline experiment, and we adopt their held-out BRO as a development set so the comparison uses the same holdout. A companion paper improves the within-set objective with an adapted InfoNCE loss but reports no unseen-set results (Bertram et al., 2024b). Outside *Magic: The Gathering*, Cardsformer (Xia et al., 2023) encodes Hearthstone card descriptions with a language model so that a gameplay policy generalises to unseen cards; it is the closest cross-game precedent for text-mediated transfer, applied to gameplay rather than drafting.

A claim of zero-shot transfer is only as strong as the discipline behind it, so we designed the evaluation to be checkable rather than trusted. The protocol of Section 4 was frozen in a tagged public commit before the test data existed. The test set is Marvel Super Heroes (MSH), a licensed-IP set whose public draft logs had not been published at the freeze. No training, tuning, or metric choice touches MSH pick data; the headline number comes from a single pre-registered inference pass over the first public snapshot. Every model-selection decision is made instead on three development sets that the development model never trains on.

On that development trio, DraftFM agrees with high-win-rate players' picks 54.3% of the time.[1] To put that number in context, consider BRO. A model trained directly on BRO's own picks reaches 65.6%, in effect a ceiling for a draft assistant on that set. DraftFM, which has never seen a BRO pick, reaches 74.3% of that ceiling on identical held-out picks. DraftFM also clears every published number we could find that a real deployment could have used on release day, including the expert-tuned ratings at 44.5%, zero-shot GPT-4o at 43%, and an hour-zero rarity heuristic we measure at 42.8% in a pre-freeze rehearsal on SOS. Each of those numbers comes from a different test set than ours, so we read them as reference points rather than a head-to-head ranking. On MSH itself, the frozen evaluation reports 57.0% top-1 agreement with high-win-rate players, while that same rarity heuristic collapses to 25.6% there (Section 7).

This paper makes the following contributions. It presents a cross-set model and training recipe: a two-tower architecture over frozen card features, in which a pre-registered comparison tests a 2.2M-parameter design that also reads the full set's card list against a simpler 1.6M-parameter version that drops that component; the simpler version wins and is what ships, the A-noctx variant of Section 6, and it trains on 149.4M picks in about 1.8 hours on a single consumer workstation

[1] The unweighted mean over the three sets: 48.9% on BRO, 57.5% on Teenage Mutant Ninja Turtles (TMT), and 56.5% on Secrets of Strixhaven (SOS).

Figure 1.1: Emeritus of Ideation, an actual first pick from a real thirteen-card Secrets of Strixhaven (SOS) pack with an empty pool. The per-set ceiling model (Table 5.1) ranks it above every other card in the pack by a wide margin: expected value[2] 8.61, seen in 65.1% of games it appears in, average last-seen-at pick 1.4, and >99% model dominance over the closest candidate in the pack, Quandrix Charm. SOS is a development set, so this figure illustrates the model's output rather than a headline result.

(Section 3). It introduces a pre-registered, tamper-evident benchmark for day-one draft evaluation, together with the first multi-set zero-shot numbers that use no post-release information of any kind. It reports a scaling study of zero-shot accuracy as a function of training-set count, from one set to 28, with set-composition probes and zero-shot seed bands at two rungs. It measures what the text embedding contributes when a set's names and flavor come from a licensed universe, such as Marvel, rather than from *Magic: The Gathering*'s own. And it releases code, the frozen protocol, data manifests, model weights, and the per-pick prediction files behind every number in this paper (Appendix B).

The rest of the paper is organized as follows. Section 2 describes the corpus, the skill covariates, and the held-out sets. Section 3 presents the card featurization and the model. Section 4 gives the pre-registered evaluation protocol. Section 5 reports results on the development sets, and Section 6 analyses where the transfer comes from and where it falls short. Section 7 runs the frozen MSH evaluation. An appendix covers limitations, reproducibility, and licensing.

2 Data

Corpus. Training data is the complete 17Lands public draft-data corpus (17Lands, 2026). It spans 31 expansions, from Strixhaven: School of Mages (STX, April 2021) through Secrets of Strixhaven (SOS, June 2026), and covers Premier (best-of-one) and Traditional (best-of-three) draft where published: 59 (set, format) files, roughly 160M pick rows. Each row is one human pick. It records the pack contents, the drafter's pool, the card taken, and two skill covariates described below. Files were fetched once from the public S3 bucket, following the 17Lands usage guidelines for bulk downloads, and frozen by sha256 and ETag in a data manifest with content hash `443bd25f72ee`. The manifest is released with the paper.

Two public files are structurally excluded from the corpus, and the exclusions are recorded in

[2] *Expected value* is the per-set model's own raw score for the card (higher is better; not the top-1 agreement metric reported elsewhere in this paper). *GIH%* and *ALSA* are 17Lands per-card statistics: GIH% is win rate in games where the card was drawn, and ALSA is the average pick number at which the card was last seen, where lower means drafters take it earlier. *Model dominance* is this paper's pairwise measure, sigmoid(top EV − runner-up EV).

the corpus registry. Through the Omenpaths (OM1) is a pick-two format whose rows carry two picks while its pool columns undercount them, and most of its digital-only cards do not exist on Scryfall. The Powered Cube is a curated 545-card singleton pool with no expansion structure. A third file, the Innistrad: Crimson Vow (VOW) QuickDraft file, is held out of the default corpus for a distributional reason rather than a structural one: it contains human picks made inside bot pods, with off-distribution pack dynamics. It is one of the two pre-registered flexibility-clause extras (§4) that may join F-full's training data if a dev-trio comparison shows a gain. Everything else is kept, including digital-only and remaster sets.

Skill covariates. 17Lands attaches two per-draft user covariates to every row, bucketed for privacy: the drafter's recent game count, in coarse buckets, and the drafter's win rate, in 2% increments. 17Lands does not publicly document the precise computation window of these covariates, so we use them as given, as a per-event skill signal. Training filters no one out; every valid pick is retained and tagged with its drafter's covariates, and skill enters the model only as a conditioning input (§3). The headline deployment conditioning uses a fixed skill bucket frozen before evaluation (§4), so the deployed number does not depend on any individual drafter's covariate.

Normalization. The corpus spans three schema eras and several quirks, all normalised at ETL time. The 2021 files are gzipped tarballs rather than plain gzipped CSVs, and the oldest era reports match-based rather than game-based skill buckets; we harmonise these, and the two are identical in best-of-one. Five sets are missing the first pick of each draft in the source logs, which we record as an annotation rather than an exclusion. Packs contain 13, 14, or 15 picks depending on the set, and the pack size is a model input (§3). Card names are joined to Scryfall by normalised name with an explicit alias table; unresolved names fail the build rather than featurizing as zeros.

Splits. Within each (set, format) shard, drafts are split by a hash of the draft identifier (`crc32 % 1000 < 50`, ∼5% validation). Early stopping uses only this within-training-set validation split. All headline evaluation is on picks by *high-win-rate players*: drafters whose win-rate bucket is at least 0.55 and whose game-count bucket is at least 100. We refer to this population as the *high-win-rate slice* throughout. The per-set ceiling models train and evaluate on the same filter, so ceiling comparisons are population-matched.

The held-out sets. Four sets never contribute a pick to model selection. The Brothers' War (BRO), Teenage Mutant Ninja Turtles (TMT), and SOS are the development trio (§4). All model selection keys on zero-shot accuracy over these three, computed from *F-dev*, which trains on none of them. Once selection is frozen, the headline model (*F-full*) retrains on all 31 sets, dev trio included; only Marvel Super Heroes (MSH) stays unseen by every model in the battery. MSH, the test set, is stricter still. A registry-level gate (`EVAL_ONLY`) makes every training and feature-fitting code path refuse MSH pick data outright, and at the protocol freeze its public data did not yet exist. MSH *card* features from Scryfall are permitted. That is the zero-shot contract: public card information only, no human behavioral data of any kind.

"Unseen" describes the set, not every card in it. BRO, TMT, and SOS packs include some ordinary reprints, mostly base-set staples and basic lands, that also appear in F-dev's training corpus under other sets. This confers no advantage under the contract above: a reprint has no memorized identity to exploit, only the same frozen feature vector any novel card with identical features would get. But it does mean set-level and card-level novelty are not the same claim.

3 Method

3.1 Cards as frozen feature vectors

Every card is a fixed 775-dimensional vector computed from public information alone, frozen before training.

Structured features (391-d). Parsed from the card's Scryfall record: mana value (scaled + bucketed one-hot, 10), mana-cost pips including hybrid/Phyrexian/X structure (10), colors (8), color identity (5), supertypes (3), card types (9), creature subtypes (top-128 vocabulary + an unmatched count, 129), power / toughness / loyalty each as (scaled, missing, star) triples (9), rarity (5), keyword abilities (166 slots + unmatched count, 167), layout including back-face flags for double-faced cards (10), and text-shape features: length, line count, and 24 regex flags over rules text for common functional categories (removal, card draw, counterspells, tokens, ramp, ...) (26). Double-faced and split cards take numeric blocks from the front face. Subtype and keyword vocabularies are counted over *training* sets only; a registry gate refuses to fit vocabularies on MSH, the frozen test set (§4). That guarantee covers MSH specifically. It does not mean the dev trio's cards were excluded from every vocabulary in the release (§2), so calling the dev trio "zero-shot" is a claim about training picks, not about where the vocabulary came from. The resulting featurizer manifest is content-hashed (`e1313cadd03b`) and released; held-out cards featurize through the frozen manifest, with out-of-vocabulary subtypes and keywords absorbed by the unmatched counters. Scalars are clipped to $[0, 1]$; parse failures set a missing indicator, never a silent zero.

Text embedding (384-d). The card's type line and rules text, embedded with a frozen `bge-small-en-v1.5` sentence encoder (Xiao et al., 2023) (L_2-normalised). The card's own name, and for legendary cards its short name, is masked to a placeholder token before embedding, so the encoder cannot key on names it may know from pretraining. This is a deliberate handicap, and it matters for licensed-IP sets (§6). Parenthetical reminder text is kept, since for a brand-new mechanic it is the only definition the model will ever see. Back faces are appended for double-faced cards.

3.2 Architecture

DraftFM is a two-tower scorer with 2.2M parameters (Figure 3.1). Its two design invariants operationalize the zero-shot contract (§2):

1. **No set identity.** There is no set embedding and no per-card parameter anywhere. The only way the model can know what set it is drafting is a *set-context tower* that reads the set's full card list, plus the drafter's accumulating pool. "Blue is the aggressive color in this set" must be inferred from the card list, not memorized as a set identifier. Whether the tower itself turns out to be necessary for that inference, or whether the invariant holds just as well without it, is addressed later in this section.

2. **Skill conditions the scorer, not the cards.** Card semantics are skill-invariant; the pick policy is not. Skill-bucket embeddings enter only the context pathway. Serving conditions on a fixed high-skill bucket; training sees every pick tagged with its drafter's bucket.

A shared *card encoder* (LayerNorm → 775–512–256 MLP, GELU) maps feature vectors to d=256 embeddings. Because training batches are homogeneous in (set, format), the encoder runs once per

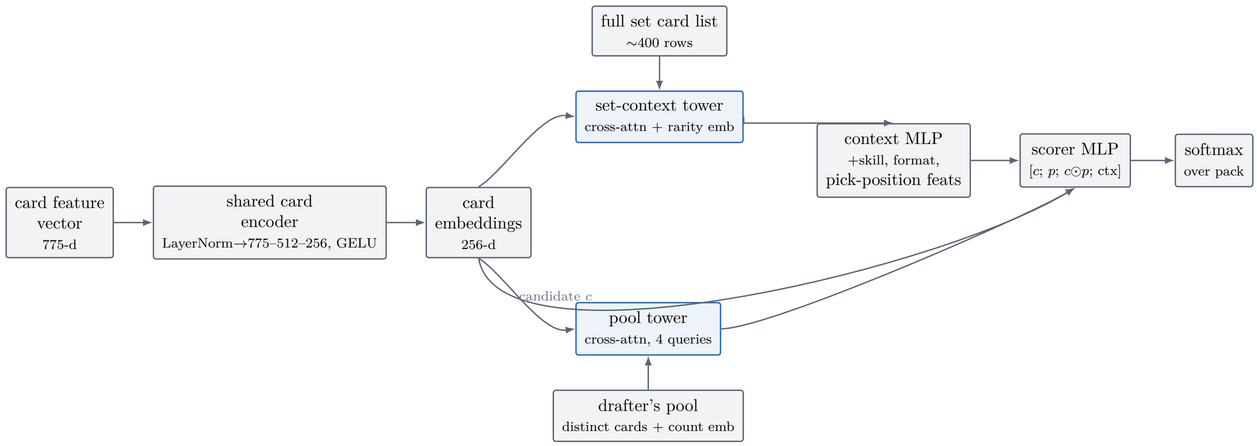

Figure 3.1: The two-tower scorer (§3.2). A shared card encoder embeds every card once per (set, format) step; a pool tower summarises the drafter's held cards, a set-context tower summarises the full card list, and a per-candidate scorer combines the candidate embedding c, the pool summary p, their elementwise product, and the context vector into a softmax over the pack.

step over the set's full card list (∼400 rows) and everything downstream gathers rows from that table. The *pool tower* represents the drafter's pool as the set of distinct cards held, each embedding translated by a learned count embedding (counts capped at 8), pooled by cross-attention from four learned queries (4 heads); an empty pool attends over a learned null token. The *set-context tower* applies the same query-pooling to the entire card list (with a rarity embedding added) to produce the set summary. A context MLP combines the set summary with skill and format embeddings, seven pick-position features (pack/pick indices, pool size), and four set-shape scalars (card-list size, picks-per-pack indicator) into a 64-d context. Each candidate c in the pack is scored by an MLP over $[\,c;\ p;\ c \odot p;\ \text{ctx}\,]$, where p is the pool summary and $c \odot p$ the elementwise interaction; a softmax over the pack's real slots gives the pick distribution.

We test the set-context tower's contribution directly with an experiment that removes it and retrains; we call this variant A-noctx (Table 6.1, §6). It scores higher on all three dev sets. That tower-free configuration, not the one just described, is what F-full and the frozen MSH evaluation actually run. The first invariant still holds either way, since neither configuration has a set-identity parameter. Dropping the tower simply means the shipped model's only route to set-relative signal is the pool tower, each candidate's own features, and the four set-shape scalars above; it never reads the rest of the card list.

3.3 Training

One recipe governs every run in the paper. It was fixed before the scaling and variant studies, since the protocol forbids per-rung tuning. Batches of 8,192 picks are drawn from one (set, format) shard at a time. Shards are sampled proportionally to $n_{\text{picks}}^{0.5}$, an interpolation between pick-uniform sampling, which lets the largest sets dominate, and set-uniform sampling, which oversamples tiny sets. The optimizer is AdamW (β_1=0.9, β_2=0.98, weight decay 0.01) with peak learning rate 10^{-3}, a 2,000-step linear warmup, and cosine decay; the loss is cross-entropy with label smoothing 0.05, renormalised over real pack slots. The presentation budget is four epochs. Within-training validation (never the dev sets) is evaluated every 2,000 steps and training stops after three non-improving evaluations. Seed 17 throughout. Seed variance is measured at two scaling rungs (§4).

Training runs on a single Apple M3 Ultra workstation (96 GB unified memory) in PyTorch 2.12 eager mode on the MPS backend, at ∼42,750 picks/s. The full 149.4M-pick run takes about 1.8 hours, stopping at step 34,000. Training-backend correctness is gated rather than assumed; specifics are in the companion reference. Inference exports to ONNX (opset 18) and runs on a consumer x86-64 machine, CPU-only, well under a millisecond per pick.

4 Pre-registered evaluation protocol

The evaluation is frozen at a public git tag (`eval-protocol-v1.1`) that predates the existence of the test data. The tag fixes the protocol document, the metric implementations, and the battery manifest, and a hash guard refuses to run if the metric code has since drifted from the tagged version. Model weights are too large to version directly, so each battery member's weights are instead anchored by the sha256 recorded in that tagged manifest, cross-checked against the experiment ledger's commit history to confirm the artifact predates the test data. There is one evaluation round; a second round would require a new public tag and disclosure that a second round happened. One terminological note for auditors: the frozen protocol document at `eval-protocol-v1.1` uses the name "expert slice" for the population this paper calls high-win-rate players (§2); the filter is identical, so the two terms map one to one.

4.1 Test set: MSH, exactly once

MSH is the first *Magic: The Gathering* set drafted on Arena after the protocol freeze. It is a licensed-IP (*Universes Beyond*) set, which makes the test deliberately harder (§6). The protocol defines the test snapshot as the first public 17Lands MSH Premier file successfully downloaded on or after its publication, frozen immediately by sha256 and ETag and never re-downloaded. Until the frozen evaluation has run, no training, tuning, metric-peeking, or feature-vocabulary fitting may read MSH pick data. MSH *card* features alone are legal, per the zero-shot contract of §2. The snapshot must pass quality gates: modern schema, Premier rows present, at least 2,500 drafts from the high-win-rate slice, and $\geq 99\%$ of pack card names joining to card features. On a volume failure the single pre-declared contingency is to wait exactly one snapshot cycle. The evaluation must execute within 72 hours of the gates passing: one inference pass per battery member over the frozen snapshot, cached as per-pick prediction files from which every statistic is computed. Models are never re-run during analysis. The execution of this protocol, and its outcome, are reported in §7.

Two populations are scored. The high-win-rate slice (§2) is the headline; the all-users slice is secondary. Two conditioning configurations are pre-registered. *Deployment mode*, the headline, conditions on a fixed top skill bucket selected on the dev sets and frozen into the battery configuration. *Human-model mode* conditions on each drafter's true bucket and is scored on all users. Forced picks, the final pick of a pack, are included in headline numbers, matching the per-set ceilings and prior work; non-forced variants are secondary rows.

4.2 Development discipline

One number governs every architecture, hyperparameter, text-encoder, and conditioning decision in this paper: mean top-1 agreement, on the high-win-rate slice in deployment mode, averaged unweighted over three sets (BRO, TMT, SOS), with ties broken by mean log-loss. That number comes from a single development run, *F-dev*, trained on every set except those three and MSH.

We chose that trio for three separate reasons. BRO is the same holdout used by the only published multi-set, feature-based zero-shot number (Bertram et al., 2024a), so our result is directly comparable to theirs. SOS is the set where our own per-set model is most mature, giving the strongest available ceiling to measure against. And TMT is a licensed-IP set, so it rehearses the same distribution shift MSH will present. Early stopping never looks at these dev metrics (§3.3), so the curve reported later in §5 is not selected on the same axis it is plotted against.

4.3 Metrics and statistics

The primary metric is top-1 agreement with the human pick, on the high-win-rate slice in deployment mode. Several secondary metrics accompany it: top-3 agreement; all-users top-1 in human-model mode; the *normalized score*, which divides the model's zero-shot top-1 by the per-set ceiling's top-1 on the same picks, where the ceiling is a model trained directly on the target set and aligned pick-by-pick on its validation split; mean log-loss; top-label expected calibration error, computed over 15 equal-mass bins with a temperature fit on the dev trio only, frozen into the battery before T0, and never fit on the test set itself (§6); per-(pack, pick) accuracy curves against the per-cell random floor; and *late-draft retention*, the same zero-shot-to-ceiling ratio restricted to picks 8 and later of each pack, where pool context dominates.

All intervals are cluster bootstraps over drafts, not picks (B=2,000, fixed seed, percentile 95% CIs); paired comparisons share resample indices. Clustering matters little, but the choice is grounded empirically. The measured intraclass correlation of pick correctness within drafts is $\rho = 0.0099$ (per-set SOS validation; 1,662 drafts, 69,803 picks), a design effect of $\sim$1.4 at $\sim$42 picks per draft. That gives CI half-widths of about $\pm$0.36pp at the 2,500-draft gate, under a binomial approximation at p=0.5. Seed variance is measured with three seeds at two scaling rungs, and set-composition variance with two probe rungs (§5).

4.4 The battery

Frozen by artifact hash before the test snapshot: (1) **F-full**, the final recipe trained on all 31 sets, the headline model; its dev-set numbers are meaningless (it trains on them) and are never quoted. A pre-registered flexibility clause lets two optional extras (the Powered Cube file and VOW QuickDraft) join F-full's training data if and only if a dev-trio comparison shows a gain; the decision is frozen with the battery and reported either way. (2) **F-dev**, the development model described above, which doubles as the top scaling rung. (3) **Scaling rungs** at 1, 2, 4, 8, 16, and 28 training sets (nested, in release order) plus two composition probes, all with the fixed recipe and a shared step cap. (4) **Variants**: A-notext, which removes the text embedding and keeps structured features only, and A-noUB, which removes the three licensed-IP training sets, both for the shift analysis of §6. (5) **Baselines**: random; an hour-zero rarity-color heuristic; and three asterisked post-release baselines (community site ratings at day $\sim$2, and two 17Lands-statistics pickers) that bound what cheap post-release information buys. (6) **Post-day-one rows**, frozen now and executed only after the zero-shot evaluation: a per-set ceiling trained on the frozen MSH snapshot with the stock per-set recipe, and F-full fine-tuned on the MSH training split with hyperparameters pinned from dev-trio trial runs.

4.5 Claims bands

To keep interpretation out of our own hands, three mutually exclusive claims paragraphs (for headline results below 45%, between 45 and 55%, and above 55%) were drafted before any test number existed and are carried in the paper source; the matching one is selected verbatim when the frozen evaluation reports (§7). In all bands the secondary claims are the same: the normalized-vs-ceiling score, the shape of the scaling curve, the analysis of what the text embedding contributes under the licensed-IP shift, and the skill-conditioning effect.

5 Results on the development sets

Zero-shot transfer on the development trio. Table 5.1 reports zero-shot transfer on the development trio. F-dev, the recipe used for architecture and hyperparameter selection, trains on 149.4M picks from 28 sets and is then evaluated on three sets it has never seen a pick from, where it agrees with high-win-rate players on 54.3% of picks. Every number in this paragraph is a dev-trio rehearsal by the protocol's own design; the paper's headline is the frozen MSH result of §7. Broken out per set, DraftFM reaches 48.9% / 57.5% / 56.5% on BRO / TMT / SOS. For comparison, a model trained directly on each of those sets, the ceiling, reaches 65.6% / 71.0% / 70.2%. DraftFM's pre-registered normalized score, its accuracy as a fraction of that ceiling on identical held-out picks per §4, is 74.3% / 81.3% / 80.2%, with dev-mean 78.6%. A simpler, unaligned version of the same ratio, taken over the full slice rather than the smaller aligned subset, corroborates this on a much larger sample: 74.5% / 81.0% / 80.5%, dev-mean 78.7%. That unaligned ratio is not itself the pre-registered metric, since the aligned version is capped by the size of the ceiling model's own held-out split. Excluding forced picks lowers the three numbers to 45.2% / 54.1% / 53.2%. The deployment-mode skill bucket selected on dev is the 0.66 win-rate bucket.

Reading the BRO row. BRO is the holdout on which Bertram et al. (2024a) report 55.4%, the strongest published unseen-set number. They reach it with card features, learned image embeddings, and 16 usage statistics computed from human play after the set's release, scored on all players rather than a win-rate-filtered population. Our 48.9% on the high-win-rate slice, 48.4% on all players as the population-matched comparator, is below that number, but the two are not a fair fight: their number assumes information that does not exist on release day. The fair comparison is against what the same paper achieves without post-release usage statistics, 33.6–35.6% on a single set, and every other day-one-feasible baseline in Table 5.2. Against that day-one bar, multi-set training on public card features closes most of the distance to the usage-statistics number while using only release-day information.

BRO is also the hardest set in the trio for transfer: its packs mix in a 63-card retro-artifact bonus sheet, examined directly in §6. Why TMT, the trio's licensed-IP set, nonetheless transfers as well as SOS, and why BRO lags both, is analysed there too. Figure 5.2 shows a real P1P1 bomb from TMT scored the same way. For sets that are in the training corpus but have no dedicated per-set model, the deployed system falls back to F-full's scoring; Figure 5.3 shows what that fallback looks like today for LTR, LCI, and DMU.

Baselines. Table 5.2 reports the baselines we measure ourselves; published numbers from prior work are situated in Table 1.1 in the introduction. The table's rows are scored on the frozen MSH snapshot; before the freeze, the same two hour-zero baselines were rehearsed on SOS, where random picking scored 23.2% and a heuristic that prefers lower rarity for playables and follows the pool's colors reached 42.8%. That rehearsal figure lands just below GPT-4o's 43% (Bertram, 2025) and close to the expert-tuned 44.5% of Ward et al. (2021), a useful caution about how much of drafting, on an in-universe set, is surface regularity. The same heuristic does far worse on MSH itself (25.6%, discussed in §7), so that caution cuts both ways. DraftFM's zero-shot numbers clear every day-one-feasible baseline on each of the three dev sets. The three asterisked post-release baselines are

Table 5.1: Zero-shot top-1 agreement (%) with high-win-rate players, deployment mode, forced picks included; 95% cluster-bootstrap confidence intervals over drafts in parentheses. Sets: The Brothers' War (BRO), Teenage Mutant Ninja Turtles (TMT), Secrets of Strixhaven (SOS), and Marvel Super Heroes (MSH). F-full's dev-trio cells are never quoted, since it trains on those three sets.

Model	BRO	TMT	SOS	Dev mean	MSH
DraftFM (F-dev, zero-shot)	48.9 (48.8, 49.0)	57.5 (57.3, 57.7)	56.5 (56.4, 56.6)	54.3	56.1 (55.9, 56.3)
DraftFM (A-noctx, selected dev recipe)	50.0 (49.9, 50.1)	57.7 (57.5, 57.9)	56.6 (56.5, 56.7)	54.8	–
DraftFM (F-full, zero-shot)	–	–	–	–	57.0 (56.8, 57.2)
Per-set ceiling	65.6	71.0	70.2 (69.8, 70.6)	68.9	TBD
F-dev / ceiling (unaligned)	0.745	0.810	0.805	0.787	TBD

Figure 5.1: Steel Seraph, BRO, the dev trio's hardest transfer set (§6). Real P1P1 expected value 9.59, the 99.4th percentile of BRO's card pool by the deployed per-set model's own ranking; the next two candidates in the pack, Skystrike Officer and Tyrant of Kher Ridges, rank at the 99.0th and 98.7th. Percentile, not a head-to-head split: Arena hides the pack between picks in this mode (§3.2).

run by the same pipeline on the frozen MSH snapshot after the zero-shot evaluation of §7.

Scaling. Figure 5.4 and Table 5.3 report zero-shot accuracy as training grows from one set (NEO, 5.4M picks) to the full 28-set universe (149.4M picks), under the fixed recipe. Single-set training is the regime of Bertram et al. (2024a)'s features-only anchor; the top rung is F-dev. Two extra seeds at S1 and S4 give within-training top-1 spreads of 0.677–0.682 and 0.688–0.689. Training itself is not noise-dominated at either rung; this checks in-distribution training stability and complements the zero-shot dev-mean seed spread reported below. A log-linear fit of dev-mean top-1 on $\log_2$(training picks) across the nested ladder (S1 through F-dev, 5.4M–149.4M picks) tracks the six rungs closely: slope 1.0 percentage point per doubling, $R^2 = 0.98$, with no rung off the fitted line by more than 0.4pp. But that same fit argues against reading a diminishing-returns point into the tested range, not for one. The per-doubling gain moves between 0.7 and 1.7pp/doubling, and it does so non-monotonically: lowest across the S2–S8 middle of the ladder, highest at the S1–S2 start and the S16–F-dev end, rather than decaying toward the top rung.

Six error-bar-free points, on a fit whose own top rung is the model selected on this axis, are not strong evidence of log-linearity. The honest reading is narrower than it looks: no diminishing returns are visible over the two orders of magnitude tested, which is not the same as showing that returns do not saturate beyond that range.

Re-scoring those same extra-seed checkpoints on the zero-shot dev-mean the curve actually

Figure 5.2: April O'Neil, Hacktivist, TMT, the dev trio's licensed-IP set (§6), a comic-book bomb the text encoder never saw described in *Magic: The Gathering* rules-text style. Real P1P1 expected value 6.22, the 96.7th percentile of TMT's card pool.

LTR: EV 6.33, pct 99.4

LCI: EV 6.10, pct 99.5

DMU: EV 5.78, pct 99.2

Figure 5.3: Three sets with no dedicated per-set model: The Lord of the Rings: Tales of Middle-earth (LTR), The Lost Caverns of Ixalan (LCI), and Dominaria United (DMU), all in F-full's 31-set training corpus. Expected values and percentiles are computed as in Figures 5.1 and 5.2. Because F-full trained on these three sets, the figure shows the deployed fallback path a newly released set would use before a per-set model exists for it; it is not a zero-shot accuracy claim.

plots, rather than on within-training top-1, gives per-seed spreads of 0.65pp at S1 and 0.42pp at S4. That is at most about a quarter of the 2.82pp gain from S1 to S4, so the rung-to-rung climb is not a seed artifact. These are three-seed spreads at two rungs that also fold in an early-stop-step difference (the two added seeds stopped at half the reference seed's step), so they bound the curve's seed noise rather than fully characterize it, standing in at S1/S4 as a proxy for the top rung.

The two composition probes offer a natural check on whether pick count alone predicts a rung's score, or whether which sets are in it matters too. At matched set count, S2b (MOM and TDM, 14.5M picks) scores 50.2% against S2's 50.9%, despite having *more* training picks than S2's 13.3M (Table 5.3). If pick count were the whole story, S2b should have scored higher, not lower. S4b (23.5M picks) scores 51.0% against S4's 51.7%, in the direction pick count would predict, but S4 also has more picks (27.8M), so that pair does not separate the two explanations. Only S2 vs. S2b isolates set count from pick count, and on that one pair, composition beats volume: which sets a rung trains on moves the score more than how many picks they contribute. Two probe pairs is not enough to generalize beyond that.

Table 5.2: Measured baselines: top-1 agreement (%) with high-win-rate players, deployment mode. Rows marked * use post-release information and bound what cheap human data buys within days of a set's release. Published numbers from prior work, which differ in test set, information regime, and pick population, appear in Table 1.1.

Method	Information used	Test set	Top-1 agreement (%)
Random	none	MSH	23.3 (23.1, 23.4)
Rarity–color heuristic	card features (hour 0)	MSH	25.6 (25.4, 25.7)
Site-ratings heuristic (day ∼2)*	post-release statistics	MSH	TBD
ALSA-argmin*	post-release statistics	MSH	TBD
shrunk-GIH-argmax*	post-release statistics	MSH	TBD

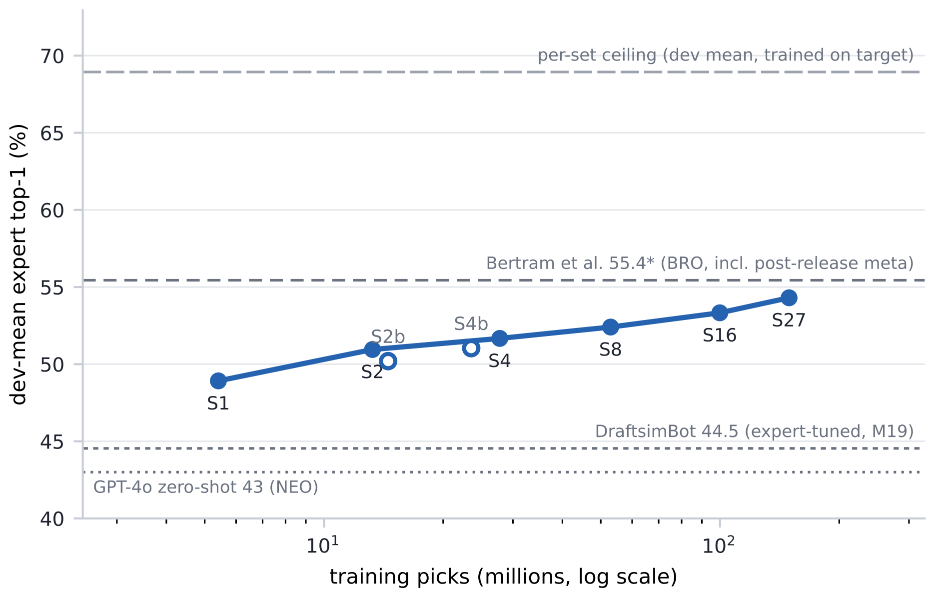

Figure 5.4: Zero-shot dev-trio-mean top-1 agreement with high-win-rate players vs. training picks (log scale). Filled points: the nested scaling ladder; open points: set-composition probes. Horizontal rules mark published anchors and the per-set ceiling mean.

6 Analysis

Table 5.3: Scaling ladder. Rung labels are frozen protocol identifiers, not set counts: *S27* names the top rung's run, which actually trains on 28 sets, a naming artifact from an earlier corpus size, kept as-is rather than renamed after the fact. Set and pick counts in the table are the real values from run records. Dev-mean is the model-selection metric of §4.

Rung	Sets	#sets	Train picks (M)	Best step	Dev-mean top-1 (%)
S1	NEO	1	5.4	4000	48.9
S2	S1 + DSK	2	13.3	18000	50.9
S2b	MOM, TDM (probe)	2	14.5	8000	50.2
S4	S2 + DMU, FIN	4	27.8	24000	51.7
S4b	STX, SNC, OTJ, TLA (probe)	4	23.5	20000	51.0
S8	S4 + STX, MOM, BLB, TLA	8	52.9	18000	52.4
S16	S8 + AFR, SNC, ONE, LTR, WOE, MKM, OTJ, EOE	16	99.9	34000	53.3
S27	full F-dev universe	28	149.4	34000	54.3

Seed variance on the top-scoring variant. Two additional seeds of A-noctx (the recipe F-full uses, §3.2) land within-training top-1 at 0.680–0.684, a 0.5pp spread around the seed reported in the table. So A-noctx's own accuracy is not a single lucky initialization. But the margins between it and the neighboring recipe-search variants are close (54.2–54.8 dev-mean, Table 6.1), meaning the full ranking among variants has not itself been seed-checked.

The flexibility-clause extras. The protocol's flexibility clause (§4) admits VOW QuickDraft and the Powered Cube into F-full's training data only if a dev-trio experiment shows a gain over A-noctx. Adding VOW QuickDraft to A-noctx (same recipe and holdout, one shard added) scores 48.9/57.6/56.0, dev-mean 54.2. That is lower than A-noctx's 50.0/57.7/56.6/54.8 on all three dev sets, and the paired-difference CI excludes zero on every set (−1.1pp BRO, CI −1.2 to −1.1; −0.1pp TMT, CI −0.26 to −0.03; −0.6pp SOS, CI −0.62 to −0.52). So the clause's own rule excludes it: F-full's frozen recipe (§3.2) does not train on VOW QuickDraft. The Powered Cube half of the clause was not tested at all, since bulk per-pick data for it is not available through the public 17Lands pipeline this paper otherwise relies on exclusively, and acquiring it by another route was out of scope. Both extras are therefore absent from F-full's training data, one by a pre-registered result that failed to show a gain and the other by an unmet data precondition, and the frozen recipe is exactly the 31-set corpus described in §2.

What the text embedding contributes on licensed universes. MSH's names and flavor come from a comic universe, not from *Magic: The Gathering*. Its text lives in the sentence encoder's pretraining as a different genre. This is the distribution shift most likely to break a text-dependent model on the day it matters, which is why the protocol makes a licensed-IP set (TMT) a development set: the rehearsal happens where it can still inform design. One mitigation is built in: card names are masked out of the text before embedding (§3.1), so the encoder sees mechanics, not franchises. A second design choice does not reduce the shift but makes it measurable. The comparison grid of Table 6.1 separates what the text embedding contributes on in-universe sets, SOS and BRO, from what it contributes on licensed sets, TMT and MSH.

Zero-shot, the licensed dev set turns out not to be the hard one. TMT reaches 57.5% against SOS's 56.5% (and 81.0% vs 80.5% of their respective ceilings), which suggests masking plus structured features are carrying most of the load. That could mean text simply adds little on licensed sets, or it could just mean TMT is an easier format. The grid distinguishes between the two:

Table 6.1: The variant grid: zero-shot top-1 (%) with high-win-rate players, point estimates (per-cell 95% CI half-widths are $\sim \pm$ 0.1–0.2pp on the dev sets, Table 5.1; paired-difference CIs accompany the deltas in the text). A-noctx, A-proportional, and A-topfilter are architecture variants tested against the Full baseline before the deployed architecture was fixed. A-notext and A-noUB are the two pre-registered comparisons (§4) analysed below: A-notext drops the 384-d text embedding from the Full recipe, and A-noUB removes the three licensed-IP training sets from A-noctx (the winning architecture, §3.2), so its clean comparator is the A-noctx row, not Full. The (variant $\times$ set) grid supplies the dev-side measurement of what the text embedding contributes on licensed-universe sets (per-set deltas below), computed before MSH is seen.

Variant	BRO	TMT	SOS	Dev mean	MSH
Full (F-dev recipe)	48.9	57.5	56.5	54.3	56.1
A-notext	45.8	55.9	53.1	51.6	54.7
A-noctx (winning)	50.0	57.7	56.6	54.8	56.2
A-proportional	50.2	56.8	55.7	54.2	55.6
A-topfilter	49.8	57.3	56.5	54.5	56.2
A-noUB	49.2	57.5	54.9	53.9	55.0

dropping text costs TMT only 1.6pp (CI 1.5–1.7pp) against its Full comparator, less than BRO's 3.1pp (CI 3.0–3.2pp) or SOS's 3.4pp (CI 3.3–3.5pp). Text contributes least on the licensed-IP set, which favors the masking-plus-structured-features explanation over TMT simply being an easier format. The UB-penalty (removing the three licensed-IP training sets from A-noctx) tells a consistent story: 0.8pp on BRO (CI 0.7–0.9pp), 0.2pp on TMT (CI 0.09–0.33pp), 1.7pp on SOS (CI 1.6–1.7pp). Losing licensed-IP training data costs the licensed-IP dev set least, not most, and even TMT's small penalty is a real, paired-CI-supported effect, not noise dressed up as a trend. If the test set is harder than the trio average, the pre-registered framing stands: a licensed-IP test makes the zero-shot claim more credible, not less.

Why BRO transfers worst. BRO sits 8.1pp below the mean of TMT and SOS, at only 74.5% of its ceiling. Of three candidate explanations, the bonus sheet looked, at first pass, like the most direct: packs offering at least one of the 63-card retro-artifact bonus sheet score 40.6% zero-shot, while packs that don't score 55.5%, a 14.8pp raw gap on 44% of BRO's picks.

That gap turns out to be mostly a different artifact in disguise. BRO's booster carries exactly one bonus-sheet card, so a pack can only "have the bonus card" before it has been picked away. By construction, *has_bonus* falls from 100% at pick 1 to 2.9% at pick 15 (BRO's last, forced pick): bonus-present packs average pick 4.1, bonus-absent packs average pick 9.3, and early picks are mechanically harder regardless of what the pack contains. Stratifying by pick number before comparing (Figure 6.1) collapses the gap to 3.1pp (CI 2.8–3.4), roughly 16% of BRO's shortfall against the trio, not something comparable in size to it.

The per-(pack, pick) curve shows the same family of artifact playing out differently. The raw BRO-vs-TMT/SOS gap widens through the pack, but BRO's booster carries one more card than TMT/SOS's (a real extra forced pick, itself a consequence of the bonus sheet enlarging the pool), so late raw picks are not apples-to-apples across sets. Correcting for the random floor at each pick reverses the trend. Artifact-heavy color-signal weakening and 2022-era recency remain plausible explanations but are not separately isolated by this grid. With the bonus-sheet slice's real contribution now measured at 16%, rather than assumed to explain most of the shortfall, more of BRO's gap is left for those factors, and for anything else this grid does not cover, to account for.

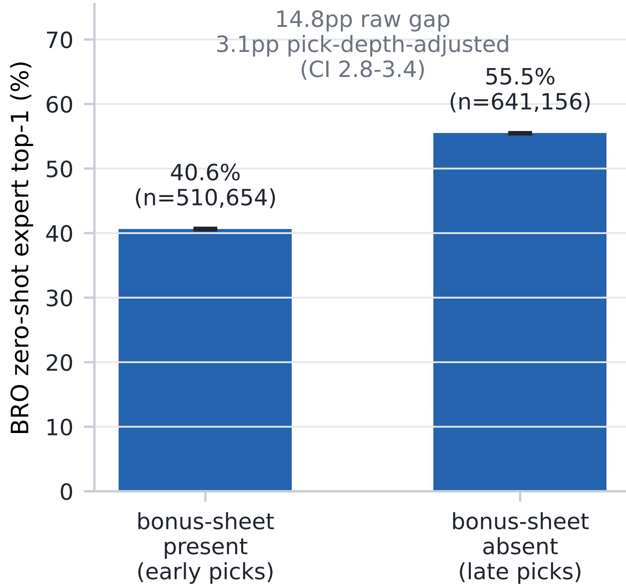

Figure 6.1: BRO zero-shot top-1 with high-win-rate players, split by whether the pack offered at least one of the 63 BRR retro-artifact bonus-sheet cards (44% of BRO's picks do). The raw 14.8pp gap is mostly a pick-depth confound (labeled above the bars): *has_bonus* tracks early-vs-late pick number by construction, and stratifying by pick number before comparing collapses the gap to 3.1pp (95% CI). Error bars on the raw bars are 95% cluster-bootstrap CIs over drafts, computed from cached zero-shot predictions.

Skill conditioning. Conditioning is a scorer-side input (§3.2), so one model serves any audience. In human-model mode (§4) the model scores 48.4% / 57.8% / 56.8% on BRO / TMT / SOS: within a point of the high-win-rate deployment numbers, consistent with skill moving picks less than card quality does. Whether MSH shows the same pattern is a question for the frozen pass (§7).

Calibration. Zero-shot confidence falls well short of within-set calibration. Top-label ECE is 0.102 / 0.071 / 0.093 on the dev trio, against 0.014 for the within-set SOS ceiling: roughly a five- to seven-and-a-half-fold gap. The protocol permits one calibration decision: a temperature fitted on dev only and frozen into the battery, never fitted on MSH. That temperature is 1.28, chosen by minimizing mean dev-trio log-loss over a grid rather than ECE directly, since log-loss is the smooth, proper scoring rule this protocol's other dev decisions already use. It cuts the dev-trio mean ECE from 0.089 at the unscaled T=1 to 0.025, closing most of the gap above. This temperature is fit on the logits of F-dev, the architecture-search model that still has the set-context tower (§3.2), and applied unchanged at test time to F-full, the tower-free recipe that actually scores MSH. F-full has no held-out set of its own to fit a temperature against without touching MSH, which is why the temperature crosses architectures this way. How that crossed-over temperature fares on MSH itself is reported with the frozen results (§7).

Late-draft retention. Later picks are where a set-agnostic model could plausibly collapse to rate-the-card, since pool signal has taken over from the card alone. The pre-registered statistic, late-draft retention (§4), is 81.8% / 87.8% / 89.2% on BRO / TMT / SOS (dev-mean 86.3%), higher, not lower, than the corresponding all-pick ratios (74.5% / 81.0% / 80.5%, §5). That raw ratio is not, on its own, evidence that the pool tower is adding signal late. Picks 8+ carry roughly

Figure 6.2: The low-conviction end of the model's output: a real thirteen-card SOS pack (empty pool) where Sundering Archaic and Flow State score a near-exact tie (EV 1.41 vs. 1.41, dominance split 50/50) – the third candidate, Pursue the Past, trails at 27% dominance against the pair. Ties this close are exactly the regime the top-label ECE below is measuring: the model's stated confidence and its actual reliability at low conviction. Computed by the deployed per-set model.

twice the forced-pick rate of the draft as a whole (forced picks score 1.0 by construction on both sides of the ratio) and a substantially smaller mean pack size (4.0–4.5 cards vs. 7.3–8.0 overall). Restricting to genuine choices, pack size ≥ 4, excluding the trivial tail, puts late-pick top-1 *below* the all-pick figure on all three dev sets, the opposite direction from the retention ratio. The claim the data does support is narrower: the model does not collapse to chance late in the draft. The per-(pack, pick) curves against the per-cell random floor show this directly, and Figure 7.1 in §7 plots them for all four sets. Excluding forced picks, zero-shot top-1 never comes closer than 20.5 (BRO) / 32.5 (TMT) / 28.4 (SOS) percentage points to chance at any point in the draft, including its latest picks.

7 The frozen MSH evaluation

Every result reported so far comes from public sets, and every modeling decision in this paper was made against the three development sets alone. While the preceding sections were being written, Marvel Super Heroes (MSH) had no public pick data at all: the set had not yet been drafted on Arena when the evaluation protocol froze. This section reports what happened when that data appeared.

The freeze. The protocol, its metric implementations, and the battery manifest are fixed at the public git tag `eval-protocol-v1.1` (§4). Each battery member's weights are anchored by a sha256 hash committed to the experiment ledger before the test snapshot existed, so the git history itself timestamps the claim that no model saw MSH. The first public 17Lands MSH file was frozen at download by its sha256 and ETag and never re-downloaded. There is one evaluation round; running a second would require a new public tag and a disclosure that a second round happened.

The single pass. One inference pass per battery member over the frozen snapshot produced cached per-pick prediction files, and every MSH statistic in this paper is computed from those files; no model has been re-run on MSH since. The headline: F-full agrees with high-win-rate players' picks on 57.0% of MSH picks, top-1, deployment mode (confidence interval in Table 5.1). The pre-registered interpretation below was drafted before that number existed and selected by it (§4).

Interpretation. The headline number exceeds the strongest published unseen-set result, the 55.44% of Bertram et al. (2024a), while using strictly less information (no post-release usage statistics, no card images) on a licensed-IP set with out-of-universe flavor (card names are masked before embedding, §3.1) – a shift the dev-trio evidence suggests text-masking mitigates but does not

Table 7.1: Top-1 agreement (%) by drafter win-rate band (bottom below 0.50, middle 0.50–0.55, top 0.55 and above), deployment mode, on the three development sets and MSH, all computed from cached predictions.

Drafter band	BRO	TMT	SOS	MSH
Bottom (win rate < 0.50)	47.8 (47.6, 47.9)	57.4 (57.1, 57.6)	56.6 (56.5, 56.7)	56.4 (56.2, 56.5)
Middle (0.50–0.55)	48.4 (48.3, 48.5)	58.1 (57.9, 58.3)	57.0 (56.9, 57.1)	56.9 (56.7, 57.1)
Top (≥ 0.55)	48.8 (48.7, 48.9)	57.8 (57.7, 58.0)	56.8 (56.7, 56.9)	57.1 (56.9, 57.2)

eliminate (§6). The BRO development row provides the same-holdout anchor against that prior. Zero-shot drafting of an unseen set is, on this evidence, better served by scale over public card features than by any published use of post-release data.

The baselines did not survive the trip. The same pass scored the pre-registered hour-zero baselines on MSH, and the result sharpens the day-one story. Random picking scores 23.3%, in line with its 23.2% pre-freeze rehearsal on Secrets of Strixhaven. The rarity-and-color heuristic, which had reached 42.8% on that rehearsal, collapses to 25.6% on MSH: barely above random (Table 5.2). Whatever surface regularity the heuristic exploited on an in-universe development set did not transfer to the licensed set, while the learned model held its level. The gap between DraftFM and the best hour-zero alternative is therefore widest exactly where day-one help matters.

Secondary detail from the same pass. The cached prediction file carries the pre-registered secondary statistics as well. Top-3 agreement is 87.7%, and mean log-loss is 1.132 nats. Top-label ECE, at the temperature fitted on the development trio and frozen before the snapshot existed (§6), is 0.005: tighter than any of the three dev-trio figures rather than between them. We do not have a confirmed explanation for that on this evidence. It is not simply forced picks inflating the count of trivially well-calibrated predictions, since MSH's non-forced ECE is close to its forced-inclusive figure, so we report it without over-interpreting it. Scored on every valid pick rather than the high-win-rate slice alone, agreement is 57.3%, close to the headline. Finally, conditioning on each drafter's own skill bucket instead of the fixed deployment bucket leaves the high-win-rate number essentially unchanged (a 0.00pp difference), so the choice of conditioning bucket matters as little on MSH as it does on the dev trio (§6).

Who the model matches. Table 7.1 breaks agreement out by drafter win-rate band, using the same cached predictions. The striking feature is how little the bands differ. On MSH the model agrees with the bottom band on 56.4% of picks, the middle band on 56.9%, and the top band on 57.1%, a spread of 0.7pp that rises mildly with drafter skill; under human-mode conditioning the top and bottom bands sit at 57.2% and 57.3%. The dev sets in the same table show the same picture, with every band within about a point of its neighbors. The model conditioned to imitate strong drafters agrees slightly more with strong drafters, and it is not disproportionately reproducing weak players' mistakes, but drafter skill moves agreement far less than which set is being drafted does.

Accuracy through the draft. Figure 7.1 plots top-1 agreement at every pick position against the per-cell random floor, for the three development sets and for MSH, all from the cached predictions. The margin over the floor is the quantity to read, since later packs are smaller and forced final picks are trivially correct. The dev-side question of §6, whether a set-agnostic model collapses

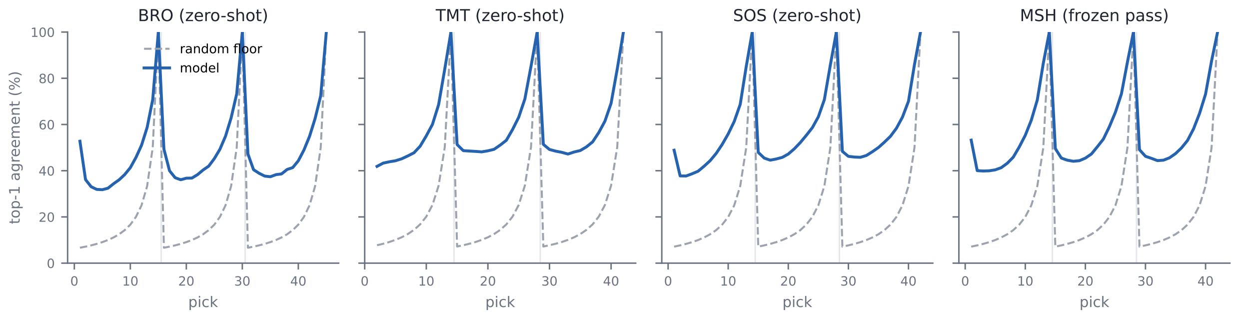

Figure 7.1: Top-1 agreement with high-win-rate players at each pick position, against the per-cell random floor, on the three development sets and MSH. Forced final picks are trivially correct, which produces the end-of-pack spikes.

toward rate-the-card once pool context dominates, can be read off the MSH panel the same way it was read off the trio.

What remains open. Two pre-registered post-day-one rows, a per-set ceiling trained on the frozen MSH snapshot and F-full fine-tuned on MSH's own picks, are deliberately not yet run, so the normalized score and late-draft retention on MSH remain open for a follow-up release ([PENDING: MSH per-set ceiling: normalized score and late-draft retention]).

Acknowledgments

This work would not exist without 17Lands, whose public datasets (17Lands, 2026) are the training and evaluation corpus, and whose data-sharing policy makes reproducible drafting research possible. Card metadata is provided by Scryfall (Scryfall, 2026). *Magic: The Gathering* is the property of Wizards of the Coast; this is unofficial research, unaffiliated with and unendorsed by Wizards of the Coast, 17Lands, or Scryfall.

References

17Lands. 17lands public datasets. `https://www.17lands.com/public_datasets`, 2026. Data from 17Lands.com, CC BY 4.0.

Timo Bertram. UrzaGPT: LoRA-tuned large language models for card selection in collectible card games, 2025. URL `https://arxiv.org/abs/2508.08382`.

Timo Bertram, Johannes Fürnkranz, and Martin Müller. Learning with generalised card representations for "magic: The gathering". In *Proceedings of the IEEE Conference on Games (CoG)*, 2024a. URL `https://arxiv.org/abs/2407.05879`.

Timo Bertram, Johannes Fürnkranz, and Martin Müller. Contrastive learning of preferences with a contextual InfoNCE loss, 2024b. URL `https://arxiv.org/abs/2407.05898`.

Daniel J. Brooks. statistical-drafting: Neural-network draft models for Magic: the Gathering. `https://github.com/danieljbrooks/statistical-drafting`, 2024. Deployed at `https://statisticaldrafting.com`; MIT license.

Philipp Czerner. puder: a neural network drafting Magic: the Gathering. `https://nicze.de/philipp/articles/puder/`, 2025. Code: `https://github.com/suyjuris/puder`.

Draftsim. Draftsim card ratings. `https://draftsim.com/ratings/`, 2026. Initial set ratings are hand-made by human experts.

Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*, 2019. URL `https://arxiv.org/abs/1908.10084`.

Scryfall. Scryfall Magic: The Gathering card database and API. `https://scryfall.com/docs/api`, 2026.

Henry N. Ward, Daniel J. Brooks, Dan Troha, Bobby Mills, and Arseny S. Khakhalin. AI solutions for drafting in Magic: the Gathering, 2021. URL `https://arxiv.org/abs/2009.00655`.

Wannian Xia, Yiming Yang, Jingqing Ruan, Dengpeng Xing, and Bo Xu. Cardsformer: Grounding language to learn a generalizable policy in Hearthstone. In *ECAI 2023 – 26th European Conference on Artificial Intelligence*, pages 2720–2727, 2023. doi: 10.3233/FAIA230581.

Shitao Xiao, Zheng Liu, Peitian Zhang, and Niklas Muennighoff. C-Pack: Packaged resources to advance general Chinese embedding, 2023. URL `https://arxiv.org/abs/2309.07597`. Model: `BAAI/bge-small-en-v1.5`.

A Limitations

The target is agreement, not optimality. Top-1 agreement with high-win-rate players measures imitation. Skilled drafters disagree with each other (the within-set ceiling is ∼70%, not 100%), and agreeing with them is not the same as maximising win rate. A model could in principle draft better than its agreement suggests, or worse. Win-rate evaluation of zero-shot drafters is future work. Training that weights picks by drafter outcome, imitating picks that correlate with winning and down-weighting or avoiding those that correlate with losing, is likewise a natural next step we leave to future work.

Population. 17Lands users are self-selected: they run a tracker, they draft on Arena, and the high-win-rate slice further conditions on volume and success. All numbers are relative to this population; paper drafting, other platforms, and casual play are out of scope.

One evaluation round. Pre-registration buys credibility at the cost of adaptivity, since the frozen battery cannot react to surprises in the test snapshot. And a single set is a sample of one from the distribution of future sets. The dev-trio spread and the composition probes bound, but do not eliminate, how much the headline number could vary on the *next* set.

Text and language. The text encoder is English-only and frozen. Rules text is embedded as prose, so mechanics whose meaning is mostly numeric or positional may be under-represented. Name masking removes the most direct pretraining leak for licensed sets, but flavor vocabulary in the remaining text still differs by universe (§6).

Numerics. Training runs on an accelerator backend with documented silent-error classes; we gate on CPU parity and skip non-finite steps (§3.3), which guards the classes we know about.

B Reproducibility

Every result in this paper traces to a run id in an append-only ledger. Every table cell and prose number is emitted by a script from that ledger and cached evaluation outputs; no result number here is typed by hand. The ledger schema, frozen identifiers (protocol tag, manifest hashes), release contents, and script-level pointers for reproducing any specific number are in the companion implementation reference, not repeated here.

C Licensing and attribution

Draft data. Draft data is from the 17Lands public datasets (17Lands, 2026), used under CC BY 4.0 and per the project's usage guidelines (bulk S3 downloads; no API scraping).

Card metadata. Card metadata is from Scryfall (Scryfall, 2026), used per its API guidelines; Scryfall neither produces nor endorses this work.

Card images. Figures 1.1, 5.1, 5.2, 5.3, and 6.2 reproduce official Scryfall card scans (Scryfall, 2026), fetched directly from Scryfall's own hosted image URLs (fetch script and source manifest released with the companion reference). Scryfall's stated policy on card images ("Use of Scryfall Data and Images," `https://scryfall.com/docs/api`) permits this use, subject to rules we follow throughout: every image is shown full and unmodified, with no cropping, distorting, blurring, sharpening, desaturating, color-shifting, or watermarking of the card art itself. Any score or label in a figure sits beside the image, never on top of it. Because the full card image is always used (never an art crop), the copyright and artist line Scryfall prints at the bottom of the card remains visible in every figure, and nothing here implies that Wizards of the Coast produced or endorses this work. The cards illustrated are drawn from BRO, SOS, TMT, LTR, LCI, and DMU, all public, released sets (§2), never from MSH, which remains untouched until the frozen evaluation of §4.

Wizards of the Coast intellectual property. Portions of the materials used, including all card names, text, and art shown in figures, are property of Wizards of the Coast. © Wizards of the Coast LLC. This work contains unofficial Fan Content permitted under the Wizards of the Coast Fan Content Policy; it is not approved or endorsed by Wizards of the Coast. *Magic: The Gathering* and its card names and text are trademarks and copyrights of Wizards of the Coast. Licensed-IP card names and imagery appear only as printed on the official card or as data strings in sourced fields of the released artifacts, and are the property of their respective rights holders (Marvel's for MSH, the Teenage Mutant Ninja Turtles franchise's for TMT): distinct properties, both used by Wizards of the Coast under license and neither affiliated with this paper.